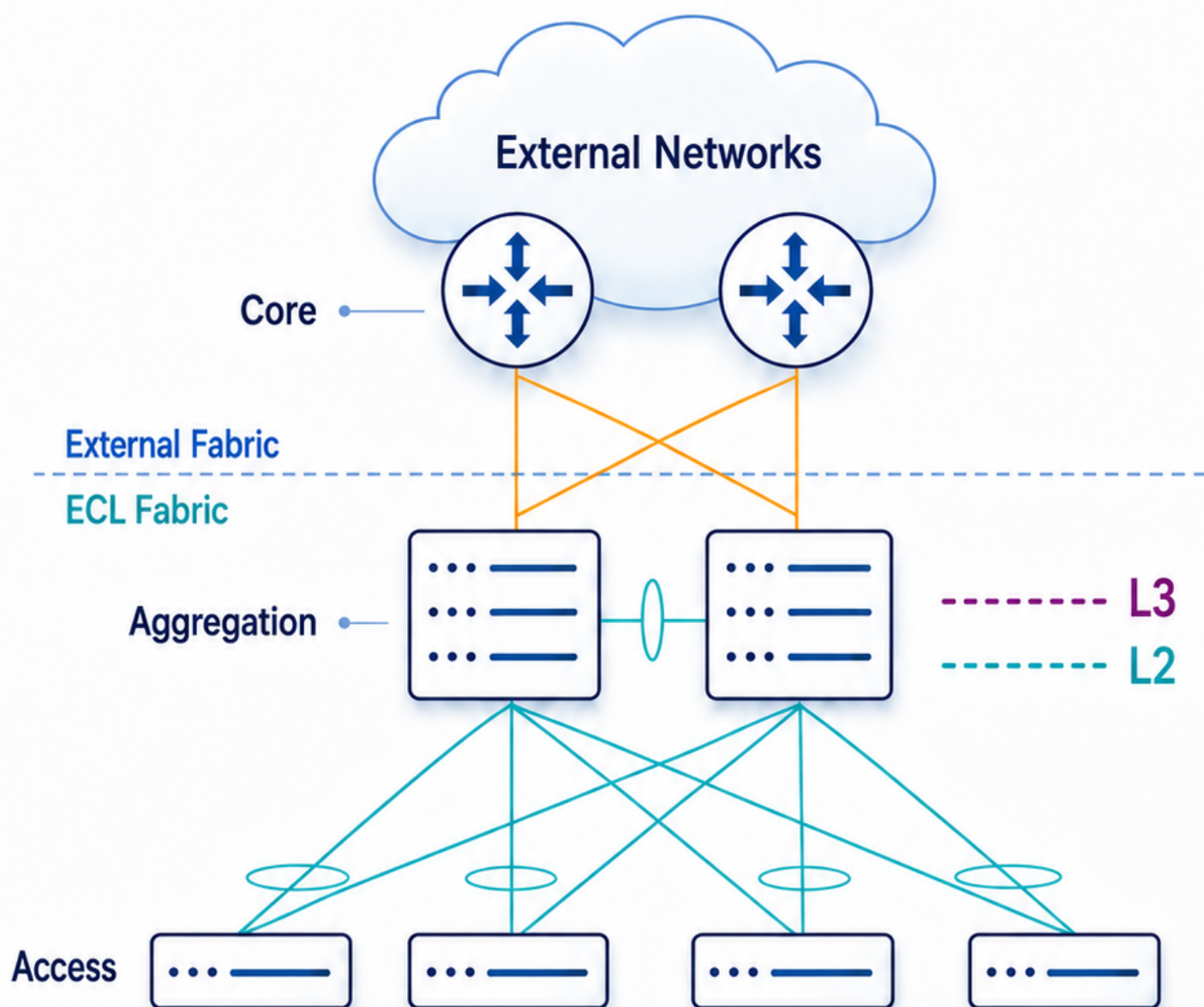


Enhanced Classic LAN with Nexus Dashboard

Hands-On Lab Guide



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Introduction

As organisations rapidly modernise their data centres and adopt newer technologies such as VXLAN EVPN, the traditional three-tier architecture comprising of the Access, Aggregation, and Core layers continues to remain relevant. Although fabric-based designs are increasingly common, the classic LAN model has not become obsolete and still exists in many production data centres supporting important workloads.

Legacy data centre environments should no longer rely on purely legacy operational methods and practices. They should benefit from the same principles of consistency, automation, and centralised management that are expected in modern infrastructures. This is where Cisco Nexus Dashboard comes into play.

Cisco Nexus Dashboard is the unified management platform for modern data center networks.

The Nexus Dashboard provides a collection of predefined fabric templates built around Cisco-recommended practices and common deployment use cases. These templates allow network configurations to be modelled centrally and deployed consistently across the required devices.

When deploying or managing a classic three-tier network with Nexus Dashboard, there are two possible approaches:

1. Building a new greenfield three-tier environment, where the fabric is created and deployed from scratch.
2. Importing an existing brownfield three-tier environment, where the current infrastructure is discovered and brought under Nexus Dashboard management.

This lab focuses on the greenfield deployment approach and uses the illustrated topology, consisting of a Core layer, dual Aggregation switches, and multiple Access switches, representing a traditional three-tier data centre LAN design.

For this lab, the Enhanced Classic LAN fabric template will be used for the Access and Aggregation layers, while the External and Inter-Fabric Connectivity fabric template will be used to configure the Core layer of the data centre.

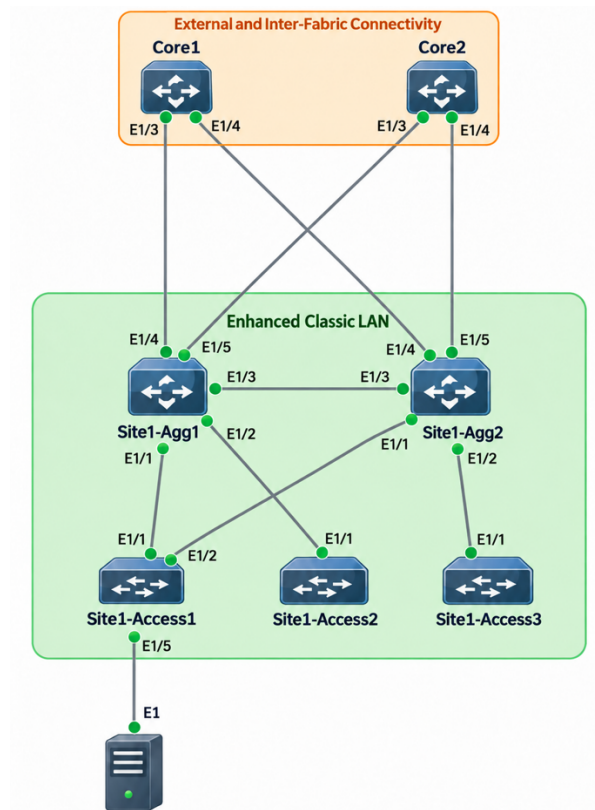
Through this greenfield example, the document demonstrates how Cisco Nexus Dashboard can be used to deploy and manage a traditional three-tier topology using a more modern and standardised operational model.

This lab will illustrate the following:

- How to create an Enhanced Classic LAN (ECL) fabric using Nexus Dashboard
- Onboarding the aggregation and access switches to the ECL.
- Configuration of end host facing ports (access)
- VRF and Network creation
- Creation of the core layer under the External and Inter-Fabric Connectivity fabric type
- VRF-Lite extension between the aggregation switches and the core switches.
- Communication between a server connected to the access switch and an external IP advertised from the core layer.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

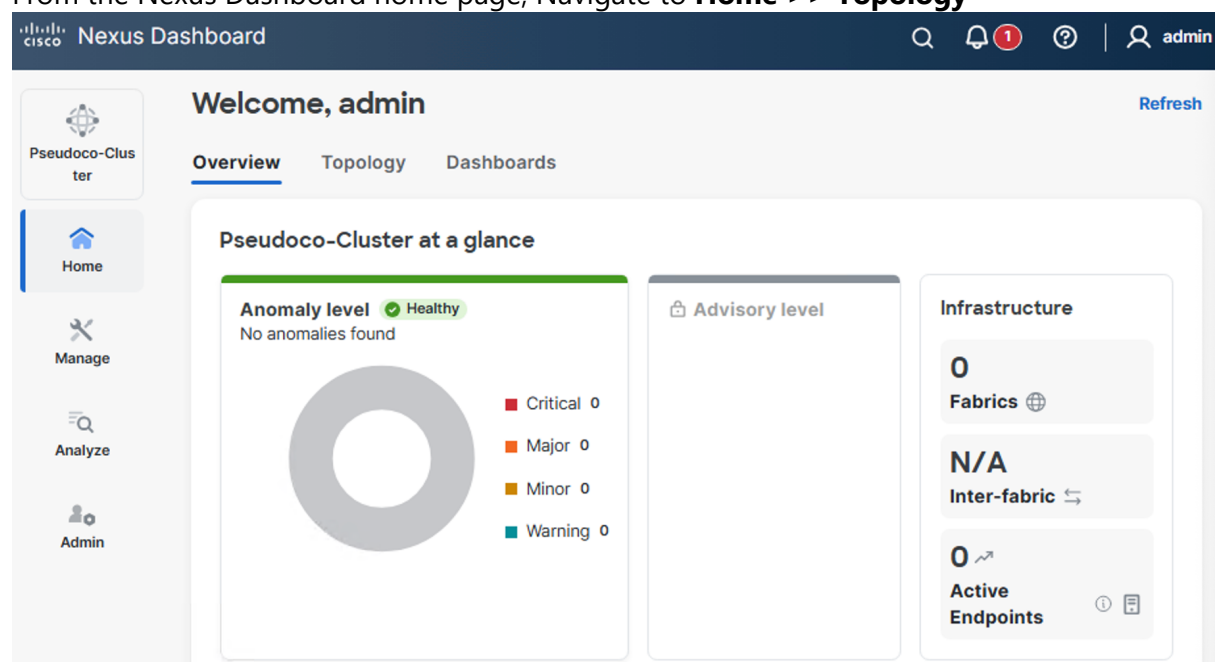
Lab Topology



Creating an Enhanced Classic LAN using ND

Step 1: Create a Fabric

From the Nexus Dashboard home page, Navigate to **Home >> Topology**



For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Click on **Actions**

The screenshot shows the Cisco Nexus Dashboard interface. On the left is a sidebar with icons for 'Pseudoc-Cluster', 'Home', and 'Manage'. The main header says 'Welcome, admin' with a 'Refresh' link. Below the header are tabs for 'Overview', 'Topology' (which is selected), and 'Dashboards'. A blue notification bar at the top of the main content area states 'Fabrics are not yet available on this setup.' Below this, there's a section titled 'All fabrics' with two tabs: 'Operational status' and 'Config-sync status'. A search bar labeled 'Filter by attributes' is present. At the bottom right of this section is a blue button labeled 'Actions' with a dropdown arrow.

Create fabric

This screenshot is similar to the previous one, but the 'Actions' button has been clicked, opening a dropdown menu. The menu contains several options: 'Layout' with a right-pointing arrow, 'Aggregate links' with a toggle switch that is currently turned on, 'Show logical links' with a toggle switch that is currently turned off, and 'Create fabric' which is highlighted in blue.

Create new LAN fabric

The screenshot shows the 'Create/Onboard Fabric' wizard. At the top left is a back arrow and the text 'Topology'. The title is 'Create/Onboard Fabric' with a 'What is a fabric?' link on the right. On the left side, there's a vertical list of steps: 1. Select a category (Create new LAN fabric), 2. Select a type, 3. Settings, and 4. Summary. The first step is active. The main content area is titled 'Select a category' and explains that the user can create a new fabric or onboard an existing one. It includes a 'Learn more' link. Below this, there's a green button with a plus icon labeled 'Create new LAN fabric'. A text box below the button explains that this option provisions a new network comprising of Cisco NX-OS, IOS-XE, IOS-XR, and/or 3rd party devices through the Nexus Dashboard.

Choose the **“Classic LAN”** type.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

✓ Select a category
Create new LAN

2 Select a type
Classic LAN

3 Settings
Default

4 Summary

5 Fabric creation

Select a type

Switches in this fabric will be configured automatically based on the option you choose.

VXLAN
 Automate a VXLAN BGP EVPN fabric for Cisco Nexus (NX-OS) and/or Catalyst (IOS-XE) switches.

Classic LAN
 Automate the provisioning of a 2 or 3-tier Traditional Classical Ethernet Network.

Fabric type Enhanced Classic LAN
 Fabric for Access-Aggregation-Core Classic LAN architectures based on Cisco best practices with Nexus 3000/7000/9000. This fabric type supports roles 'Access' and 'Aggregation'. Switches that serve as Core for such networks must be deployed in 'External and inter-fabric connectivity' fabric type.

Put in the site-specific details. This lab will use EBGp as the VRF-lite protocol.

✓ Select a category
Create new LAN fabric

✓ Select a type
Classic LAN

3 Settings
Advanced

4 Advanced settings

5 Summary

6 Fabric creation

Settings

These are the recommended settings for configuring the parameters and capabilities of the new fabric.

Configuration mode ⓘ

☐ Default ☒ Advanced

Name *

Location *

VRF-Lite protocol *

BGP ASN *

1-4294967295 | 1-65535[0-65535]

Fabric type Enhanced Classic LAN

Select the desired first hop redundancy protocol (this lab will use hsrp)

✓ Select a category
Create new LAN fabric

✓ Select a type
Classic LAN

✓ Settings
Advanced

4 Advanced settings

5 Summary

Advanced settings

The following optional settings will be deployed and/or used when deploying this fabric.

General Parameters Spanning Tree vPC **Protocols** Security Advanced Freeform

First Hop Redundancy Protocol

HSRP or VRRP

☐ **Enable Performance Monitoring**

If enabled, switch metrics are collected through periodic SNMP polling. Alternative to real-time telemetry

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

The STP Bridge Priority is set to 0 to ensure that the aggregation boxes always remain as the root bridge for the respective VLANs

Advanced settings
The following optional settings will be deployed and/or used when deploying this fabric.

General Parameters **Spanning Tree** vPC Protocols Security Advanced Freeform

Spanning Tree Root Bridge Protocol
rpvst+

Protocol to be used for configuring Root Bridge: rpvst+: Rapid Per-VLAN Spanning Tree, mst: Multiple Spanning Tree, unmanaged (default): STP Root not managed by ND. Note: Spanning Tree Settings and Bridge Configs are applicable at Aggregation layer only

Spanning Tree VLAN Range*
1-3967

Vlan range, Example: 1,3-5,7,9-11, Default is 1-3967 (Applicable only for Aggregation devices)

MST Instance Range
MST instance range, Example: 0-3,5,7-9, Default is 0 (Applicable only for Aggregation devices)

Spanning Tree Bridge Priority
0

Bridge priority for the spanning tree in increments of 4096 (Applicable only for Aggregation devices)

Spanning Tree Hello Interval
2

Set the number of seconds between generation of config bpdu, default is 2 (Applicable only for Aggregation devices)

All the fields are automatically populated based on Cisco-recommended best practice configurations, but you can update the fields if needed.

Advanced settings
The following optional settings will be deployed and/or used when deploying this fabric.

General Parameters Spanning Tree **vPC** Protocols Security Advanced F

vPC Auto Recovery Time (In Seconds)
360
(Min:240, Max:3600)

vPC Delay Restore Time (In Seconds)
150
(Min:1, Max:3600)

vPC Delay Restore Time for Access (In Seconds)
30
(Min:1, Max:3600), default is 30 if not specified

vPC Peer Link Port Channel ID
500
(Min:1, Max:4096)

☒ **vPC IPv6 ND Synchronize**
Enable IPv6 ND synchronization between vPC peers

vPC Domain Id Range
1-1000
vPC Domain Id range to use for new pairings

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Tick the **vPC Layer-3 Peer-Router Option** to ensure that if one of the aggregation switches loses connectivity to the uplink core, routing can occur across the peer link.

☒ **vPC Layer-3 Peer-Router Option**
Enable Layer-3 Peer-Router on all Aggregation Devices

☐ **Allow Selective VLANs on vPC Peer-link**
Allow VLANs associated with VRFs/Networks to be trunked on the vPC Peer-Link between Aggregation Devices

☐ **Use Specific vPC/Port-Channel ID Range**
Use specific vPC/Port-channel ID range for aggregation-access pairings

vPC/Port-Channel ID Range

Specify one vPC/Port-Channel ID range, this range is used for auto-allocating vPC/Port-Channel IDs for aggregation-access pairings

Go to the “Advanced” tab and select enable under the **Add Switches without Reload** drop-down.

By default, Nexus Dashboard will reload switches when they are added to the fabric with Preserve Config unchecked. By disabling this reload, it saves some time since the devices will not reload

Add Switches without Reload

enable

Allow switch configuration to be cleared without a reload when Preserve Config is un-checked

The Network VLAN range is the range that Nexus Dashboard will auto-assign to the Networks.

Advanced settings
The following optional settings will be deployed and/or used when deploying this fabric.

vPC Protocols Security **Advanced** Freeform Resources Manageability

Network VLAN Range

Per Switch Network VLAN Range (Min:2, Max:4094)

Aggregation-Core/Aggregation-Edge Connectivity

VRF Lite Aggregation-Core and Aggregation-Edge Router Inter-Fabric Connection Options

VRF Lite Subinterface dot1q Range

Per Aggregation dot1q Range for VRF Lite Connectivity (Min:2, Max:4093)

☒ **Auto Generate VRF Lite Configuration on Aggregation and Core/Edge**
Flag that controls auto generation of VRF Lite sub-interface and peering configuration on Aggregation & Core/Edge devices. If set, auto created VRF Lite links will have 'Auto Generate Flag' enabled

VRF Lite IP Version*

Choice of IPv4, IPv6 or both

Configure the desired VRF Lite Subnet range.

IPv4 VRF Subnet IP Range*

IPv4 address range to assign P2P Aggregation-Core connections, and peering between vPC Aggregation switches

IPv4 VRF Lite Subnet Mask Length*

IPv6 VRF Subnet IP Range

IPv6 address range to assign P2P Aggregation-Core connections, and peering between vPC Aggregation switches

IPv6 VRF Lite Subnet Mask Length*

VRF Lite VLAN Range

Review Summary and Click Submit

✓ Select a category
Create new LAN fabric

✓ Select a type
Classic LAN

✓ Settings
Advanced

✓ Advanced settings

5 Summary

6 Fabric creation

Summary

Review your selections below.

Category

Fabric categoryNew LAN fabric

Type

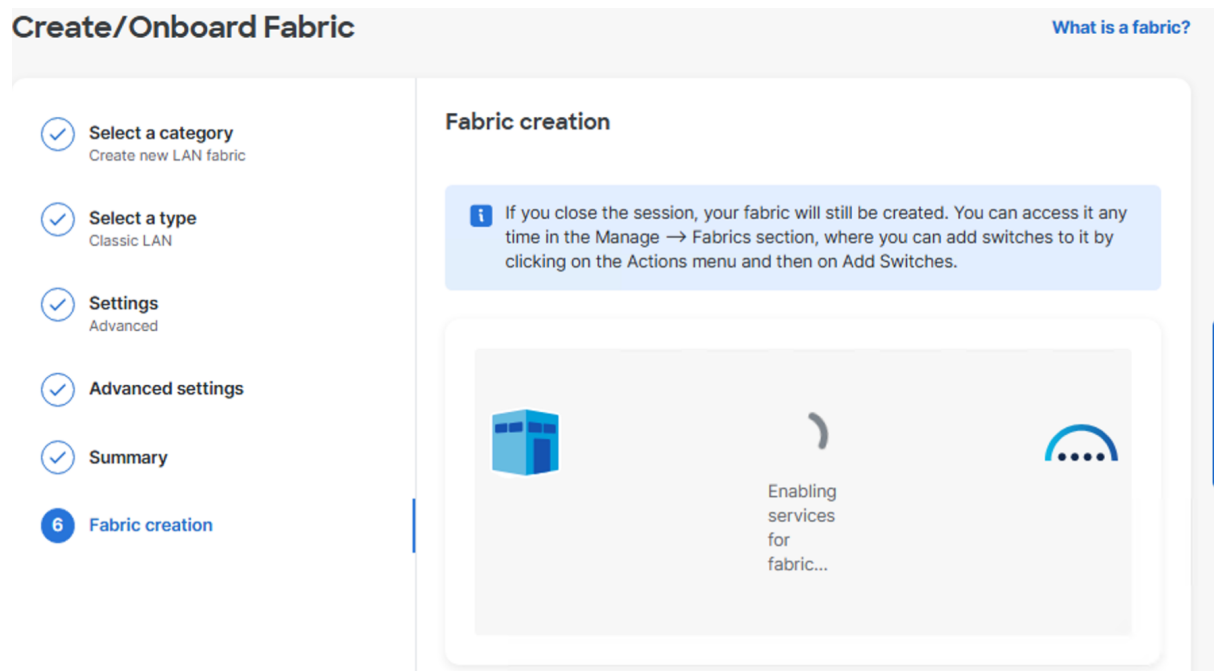
Fabric typeClassic LAN

Fabric sub-typeEnhanced Classic LAN

Settings

Name	Soweto-LAN-DC
Location	Johannesburg, ZA
License tier for fabric	Essentials
Security domain	all
VRF-Lite protocol	ebgp
BGP ASN	65002

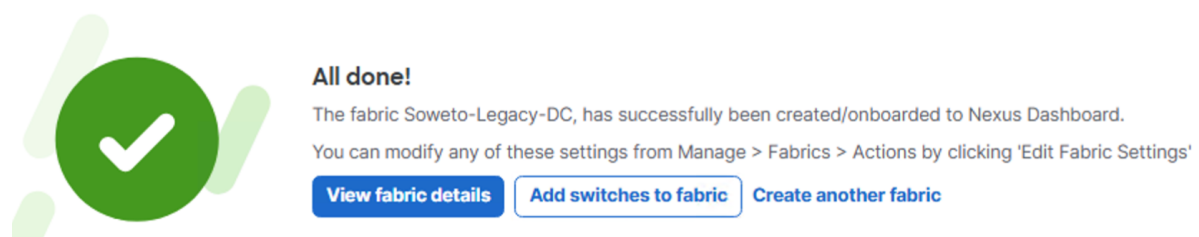
Wait for the Fabric creation process to complete.



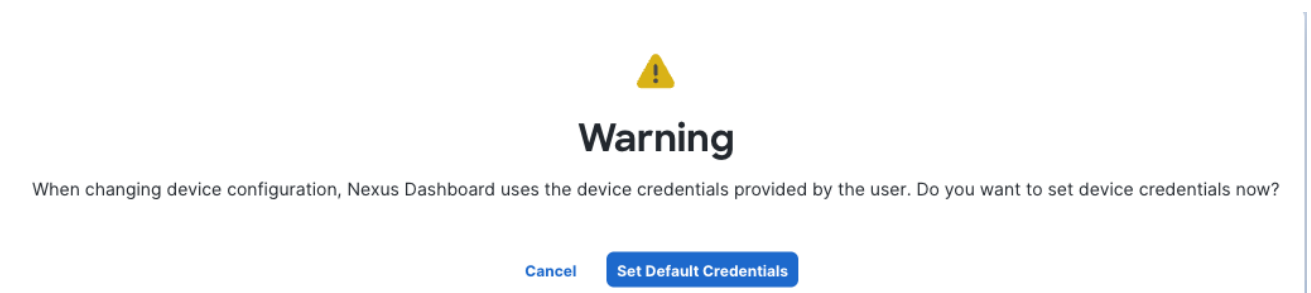
Onboard Switches to the Fabric

Next Stop is to **add Switches** to the fabric.

Click the **“Add switches to fabric”** option.



Select **“Set Default Credentials”** from the pop-up.



Enter the switch credentials that Nexus Dashboard will use to discover and manage the devices.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Set Default Credentials



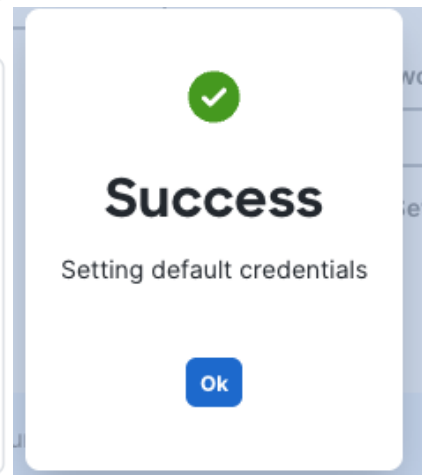
Username*

Password*

 [Show](#)

Confirm password*

 [Show](#)

☐ Robot
Robot selection allows all users with appropriate roles, to push configurations to devices without setting up device credentials.

Enter the seed IP (management switch) of one device and the other directly connected devices will be discovered via CDP (Max hop = 2). Untick Preserve config Option.

Add switches - Fabric: Soweto-LAN-DC

Switch addition mechanism*

☒ Discover

Seed switch details

Seed IP*

Ex: "2.2.2.20" or "10.10.10.40-60" or "2.2.2.20, 2.2.2.21"

Authentication / Privacy*

Username*

Password*

 [Show](#)

☐ Set as individual device write credential

Max hops*

☐ Preserve config

Unchecking this will clean up the configuration on switch(es)

Click **Discover Switches**

A warning window will pop-up. Click **"Confirm"** to proceed.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>



Warning

All switch configuration other than management, will be removed immediately after import. Do you want to proceed?

[Cancel](#)

[Confirm](#)

The discovered switches appear with their serial numbers and status, ready to be added to the fabric.

Add switches - Fabric: Soweto-LAN-DC

Switch addition mechanism*

☒ Discover

Seed switch details

Fabric
Soweto-LAN-DC

Switch
198.18.4.126

Authentication protocol
md5

Username
admin

Password

Max hops
2

Preserve config

☐ Set as individual device
write credential

☒ Disabled

[← Back](#)

Discovery results

Filter by attributes

☐	Switch name	Serial number	IP address	Model	Version	Status
☐	Core1	9RM171U1XO7	198.18.4.124	N9K-C9300v	10.6(1)	Manageable
☐	Core2	900JLZX4A1	198.18.4.125	N9K-C9300v	10.6(1)	Manageable
☒	Site1-Access1	905CRZXTNQ4	198.18.4.126	N9K-C9300v	10.6(1)	Manageable
☒	Site1-Access2	9E9K42M4KVB	198.18.4.127	N9K-C9300v	10.6(1)	Manageable
☒	Site1-Access3	9BKEW2I8YJR	198.18.4.128	N9K-C9300v	10.6(1)	Manageable
☒	Site1-Agg1	9GTE45OSVS7	198.18.4.129	N9K-C9300v	10.6(1)	Manageable
☒	Site1-Agg2	95NLDHLAI0D	198.18.4.130	N9K-C9300v	10.6(1)	Manageable

Click **Add Switches** and monitor the progress until completion

☐	Site1-Access1	905CRZXTNQ4	198.18.4.126	N9K-C9300v	10.6(1)	In progress
☐	Site1-Access2	9E9K42M4KVB	198.18.4.127	N9K-C9300v	10.6(1)	In progress
☐	Site1-Access3	9BKEW2I8YJR	198.18.4.128	N9K-C9300v	10.6(1)	In progress
☐	Site1-Agg1	9GTE45OSVS7	198.18.4.129	N9K-C9300v	10.6(1)	In progress
☐	Site1-Agg2	95NLDHLAI0D	198.18.4.130	N9K-C9300v	10.6(1)	In progress

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Nexus Dashboard imports the switches and begins provisioning them into the fabric.

Discovery results

Filter by attributes

☐	Switch name	Serial number	IP address	Model	Version	Status
☐	Core1	9RM171U1XO7	198.18.4.124	N9K-C9300v	10.6(1)	Manageable
☐	Core2	900JLZX4A1	198.18.4.125	N9K-C9300v	10.6(1)	Manageable
☐	Site1-Access1	905CRZXTNQ4	198.18.4.126	N9K-C9300v	10.6(1)	Switch added
☐	Site1-Access2	9E9K42M4KVB	198.18.4.127	N9K-C9300v	10.6(1)	Switch added
☐	Site1-Access3	9BKEW2I8YJR	198.18.4.128	N9K-C9300v	10.6(1)	Switch added
☐	Site1-Agg1	9GTE45OSVS7	198.18.4.129	N9K-C9300v	10.6(1)	Switch added
☐	Site1-Agg2	95NLDHLAI0D	198.18.4.130	N9K-C9300v	10.6(1)	Switch added

In the **Fabric Overview** window, click **Inventory** tab, and **Switches** sub-tab.

Notice how every device has been assigned the **Access** role. This is the default for the Nexus 9300 switches.

Soweto-LAN-DC

Refresh View

Inventory Connectivity Segmentation and security Configuration policies Anomalies Advisories In

Switches VPC pairs Other devices

Filter by attributes

☐	Name	IP address	Model	Configuration sync status	Role	Discovery
☐	Site1-Access1	198.18.4.126	N9K-C9300v	Not applicable	Access	Ok
☐	Site1-Access2	198.18.4.127	N9K-C9300v	Not applicable	Access	Ok
☐	Site1-Access3	198.18.4.128	N9K-C9300v	Not applicable	Access	Ok
☐	Site1-Agg1	198.18.4.129	N9K-C9300v	Not applicable	Access	Ok
☐	Site1-Agg2	198.18.4.130	N9K-C9300v	Not applicable	Access	Ok

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Select **Site1-Agg1** and **Site1-Agg2** then click **Actions > Set Role**.

Switches VPC pairs Other devices

Filter by attributes

Actions ^

Name	IP address	Model	Configuration sync status	Role	Discovery status
☐ Site1-Access1	198.18.4.126	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access2	198.18.4.127	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access3	198.18.4.128	N9K-C9300v	Not applicable	Access	Ok
☒ Site1-Agg1	198.18.4.129	N9K-C9300v	Not applicable	Access	Ok
☒ Site1-Agg2	198.18.4.130	N9K-C9300v	Not applicable	Access	Ok

- Add switches
- Configuration >
- Discovery >
- Set role
- VPC pairing
- Access Pairing
- VPC overview

In the list, choose **Aggregation** and then click **Select**. If you receive another warning message about performing a **Recalculate Config**, click **Ok**.

Switches VPC pairs Other devices

Filter by attributes

Actions v

Select Role

Search Role

Access (current)

Aggregation

Name	IP address	Model	Configuration sync status	Role	Discovery status
☐ Site1-Access1	198.18.4.126	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access2	198.18.4.127	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access3	198.18.4.128	N9K-C9300v	Not applicable	Access	Ok
☒ Site1-Agg1	198.18.4.129	N9K-C9300v	Not applicable	Access	Ok
☒ Site1-Agg2	198.18.4.130	N9K-C9300v	Not applicable	Access	Ok

The Classic LAN fabric type requires that Aggregation switches are configured in a vPC pair. To configure a vPC pair, click the checkbox next to **Site1-Agg1**, then select **Actions > VPC pairing**.

Switches VPC pairs Other devices

Filter by attributes

Actions ^

Name	IP address	Model	Configuration sync status	Role	Discovery status
☐ Site1-Access1	198.18.4.126	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access2	198.18.4.127	N9K-C9300v	Not applicable	Access	Ok
☐ Site1-Access3	198.18.4.128	N9K-C9300v	Not applicable	Access	Ok
☒ Site1-Agg1	198.18.4.129	N9K-C9300v	Not applicable	Aggregation	Ok
☐ Site1-Agg2	198.18.4.130	N9K-C9300v	Not applicable	Aggregation	Ok

- Add switches
- Configuration >
- Discovery >
- Set role
- VPC pairing
- Access Pairing
- VPC overview

Select the peer switch, *Site1-Agg2*, to pair with.

VPC pairing

Select VPC peer for Site1-Agg1

Filter by attributes			
Device	Recommended	Reason	
☒ Site1-Agg2	True	Switches are connected and have same role	
☐ Site1-Access3	False	Switches have different roles	
☐ Site1-Access1	False	Switches have different roles	
☐ Site1-Access2	False	Switches have different roles	

Recalculate and deploy

Soweto-LAN-DC Refresh View in topology Actions

Inventory **Connectivity** **Segmentation and security** **Configuration policies** **Anomalies** **Advisories** **Maintenance**

Switches **VPC pairs** **Other devices**

Filter by attributes

Name	IP address	Model	Configuration sync status	Role	Discovery status	
☐ Site1-Access1	198.18.4.126	N9K-C9300v	Not applicable	Access	Ok	
☐ Site1-Access2	198.18.4.127	N9K-C9300v	Not applicable	Access	Ok	
☐ Site1-Access3	198.18.4.128	N9K-C9300v	Not applicable	Access	Ok	
☐ Site1-Agg1	198.18.4.129	N9K-C9300v	Pending	Aggregation	Ok	
☐ Site1-Agg2	198.18.4.130	N9K-C9300v	Pending	Aggregation	Ok	

The Deployment window will display the number of configuration lines that will be added to each device.

Deploy configuration - Soweto-LAN-DC

1 Config preview **2 Deploy progress**

Filter by attributes Resync all

Switch name	IP address	Role	Serial number	Configuration sync status	Pending config	Diff	Status description	Progress	Resync
Site1-Agg1	198.18.4.129	Aggregation	9GTE45OSV S7	Out of sync	505 Lines	+306 -61	Out-of-Sync		Re
Site1-Access2	198.18.4.127	Access	9E9K42M4 KVB	Out of sync	528 Lines	+333 -63	Out-of-Sync		Re
Site1-Access1	198.18.4.126	Access	905CRZXTN Q4	Out of sync	524 Lines	+329 -62	Out-of-Sync		Re
Site1-Access3	198.18.4.128	Access	9BKEW2I8Y JR	Out of sync	528 Lines	+333 -63	Out-of-Sync		Re
Site1-Agg2	198.18.4.130	Aggregation	95NLDHLAI OD	Out of sync	505 Lines	+306 -61	Out-of-Sync		Re

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

You can review the Pending configuration (dry run) before deploying.

Pending config - Soweto-LAN-DC - Site1-Agg1

Pending config Side-by-side comparison

```
vpc domain 1
  ip arp synchronize
  peer-gateway
  peer-switch
  delay restore 150
  peer-keepalive destination 198.18.4.130 source 198.18.4.129
  auto-recovery reload-delay 360
  ipv6 nd synchronize
interface port-channel500
  switchport
  switchport mode trunk
  spanning-tree port type network
  description "vpc-peer-link Site1-Agg1--Site1-Agg2"
  no shutdown
  vpc peer-link
configure terminal
interface ethernet1/3
  channel-group 500 force mode active
  description "P0 500 (vpc-peer-link) member Site1-Agg1-Ethernet1/3 to Site1-Agg2-Ethernet1/3"
  no shutdown
configure terminal
interface port-channel1
  switchport
  switchport mode trunk
  switchport trunk allowed vlan none
  description vPC-aggregation-connected-to-access:Site1-Access1-port-channel501
```

Pending config - Soweto-LAN-DC - Site1-Access2

Pending config Side-by-side comparison

```
interface port-channel501
  switchport
  switchport mode trunk
  switchport trunk allowed vlan none
  description access-connected-to-aggregation:Site1-Agg1-port-channel501
  mtu 9216
  spanning-tree bpduguard disable
  spanning-tree guard loop
  no shutdown
configure terminal
interface ethernet1/1
  channel-group 501 force mode active
  no shutdown
```

Click **Deploy all** for the changes to be pushed down to the devices.

Monitor the Deployment Progress until completion.

Deploy configuration - Soweto-LAN-DC

Filter by attributes

Switch name	IP address	Configuration sync status	Status description	Progress
Site1-Agg1	198.18.4.129	Started	Deployment in progress.	Executed 74 / 505
Site1-Access2	198.18.4.127	Started	Deployment in progress.	Executed 176 / 528
Site1-Access1	198.18.4.126	Started	Deployment in progress.	Executed 189 / 524
Site1-Access3	198.18.4.128	Started	Deployment in progress.	Executed 177 / 528
Site1-Agg2	198.18.4.130	Started	Deployment in progress.	Executed 54 / 505

Deploy configuration - Soweto-LAN-DC

Filter by attributes

Switch name	IP address	Configuration sync status	Status description	Progress
Site1-Agg1	198.18.4.129	Success	Deployment completed.	Executed 505 / 505
Site1-Access2	198.18.4.127	Success	Deployment completed.	Executed 528 / 528
Site1-Access1	198.18.4.126	Success	Deployment completed.	Executed 524 / 524
Site1-Access3	198.18.4.128	Success	Deployment completed.	Executed 528 / 528
Site1-Agg2	198.18.4.130	Success	Deployment completed.	Executed 505 / 505

At this point, Day 0 configurations have been pushed to the devices. You have a fully functional access-aggregation setup.

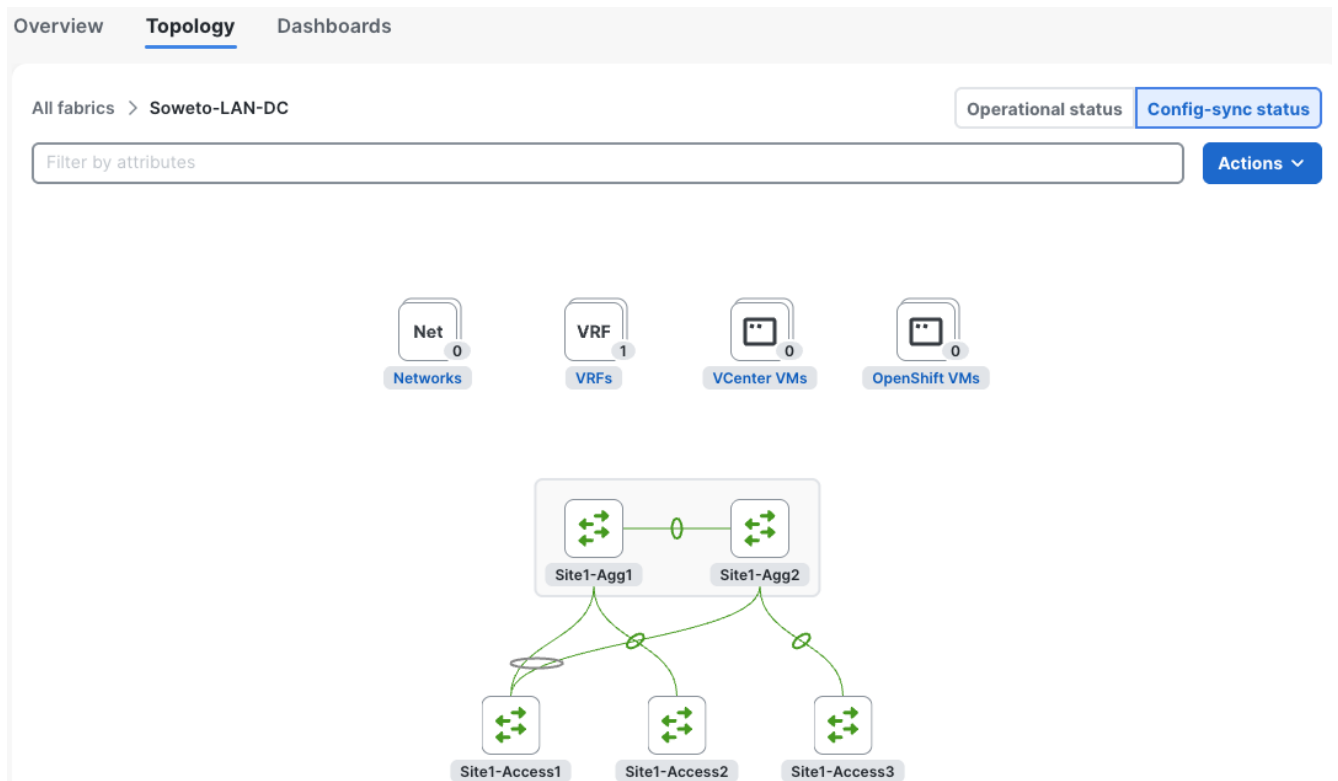
Site1-Agg1# show vpc Legend: (*) - local vPC is down, forwarding via vPC peer-link vPC domain id : 1 Peer status : peer adjacency formed ok vPC keep-alive status : peer is alive Configuration consistency status : success Per-vlan consistency status : success	Site1-Agg2# show vpc Legend: (*) - local vPC is down, forwarding via vPC peer-link vPC domain id : 1 Peer status : peer adjacency formed ok vPC keep-alive status : peer is alive Configuration consistency status : success Per-vlan consistency status : success
--	--

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Type-2 consistency status : success vPC role : secondary Number of vPCs configured : 1 Peer Gateway : Enabled Dual-active excluded VLANs : - Graceful Consistency Check : Enabled Auto-recovery status : Enabled, timer is off.(timeout = 360s) Delay-restore status : Timer is off.(timeout = 150s) Delay-restore SVI status : Timer is off.(timeout = 10s) Delay-restore Orphan-port status : Timer is off.(timeout = 0s) Operational Layer3 Peer-router : Disabled Virtual-peerlink mode : Disabled vPC Peer-link status ----- <table><tr><th>id</th><th>Port</th><th>Status</th><th>Active vlans</th></tr><tr><td>1</td><td>Po500</td><td>up</td><td>1</td></tr></table> vPC status ----- <table><tr><th>Id</th><th>Port</th><th>Status</th><th>Consistency</th><th>Reason</th></tr><tr><td colspan="5">Active vlans</td></tr><tr><td>1</td><td>Po1</td><td>up</td><td>success</td><td>success</td></tr><tr><td colspan="5">-</td></tr></table>	id	Port	Status	Active vlans	1	Po500	up	1	Id	Port	Status	Consistency	Reason	Active vlans					1	Po1	up	success	success	-					Type-2 consistency status : success vPC role : primary Number of vPCs configured : 1 Peer Gateway : Enabled Dual-active excluded VLANs : - Graceful Consistency Check : Enabled Auto-recovery status : Enabled, timer is off.(timeout = 360s) Delay-restore status : Timer is off.(timeout = 150s) Delay-restore SVI status : Timer is off.(timeout = 10s) Delay-restore Orphan-port status : Timer is off.(timeout = 0s) Operational Layer3 Peer-router : Disabled Virtual-peerlink mode : Disabled vPC Peer-link status ----- <table><tr><th>id</th><th>Port</th><th>Status</th><th>Active vlans</th></tr><tr><td>1</td><td>Po500</td><td>up</td><td>1</td></tr></table> vPC status ----- <table><tr><th>Id</th><th>Port</th><th>Status</th><th>Consistency</th><th>Reason</th></tr><tr><td colspan="5">Active vlans</td></tr><tr><td>1</td><td>Po1</td><td>up</td><td>success</td><td>success</td></tr><tr><td colspan="5">-</td></tr></table>	id	Port	Status	Active vlans	1	Po500	up	1	Id	Port	Status	Consistency	Reason	Active vlans					1	Po1	up	success	success	-				
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Site1-Agg1# show run interface port-channel1 !Command: show running-config interface port-channel1 version 10.6(1) Bios:version interface port-channel1 description vPC-aggregation-connected-to-access:Site1-Access1-port-channel501 switchport mode trunk switchport trunk allowed vlan none spanning-tree bpduguard disable spanning-tree guard root mtu 9216 vpc 1	Site1-Agg2# show run interface port-channel1 !Command: show running-config interface port-channel1 version 10.6(1) Bios:version interface port-channel1 description vPC-aggregation-connected-to-access:Site1-Access1-port-channel501 switchport mode trunk switchport trunk allowed vlan none spanning-tree bpduguard disable spanning-tree guard root mtu 9216 vpc 1																																																								
Site1-Access1# show port-channel summary Flags: D - Down P - Up in port-channel (members) I - Individual H - Hot-standby (LACP only) s - Suspended r - Module-removed b - BFD Session Wait S - Switched R - Routed U - Up (port-channel) p - Up in delay-lacp mode (member) M - Not in use. Min-links not met ----- <table><tr><th>Group</th><th>Port-Channel</th><th>Type</th><th>Protocol</th><th>Member Ports</th></tr><tr><td>501</td><td>Po501(SU)</td><td>Eth</td><td>LACP</td><td>Eth1/1(P) Eth1/2(P)</td></tr></table>	Group	Port-Channel	Type	Protocol	Member Ports	501	Po501(SU)	Eth	LACP	Eth1/1(P) Eth1/2(P)	Site1-Access2# show port-channel summary Flags: D - Down P - Up in port-channel (members) I - Individual H - Hot-standby (LACP only) s - Suspended r - Module-removed b - BFD Session Wait S - Switched R - Routed U - Up (port-channel) p - Up in delay-lacp mode (member) M - Not in use. Min-links not met ----- <table><tr><th>Group</th><th>Port-Channel</th><th>Type</th><th>Protocol</th><th>Member Ports</th></tr><tr><td>501</td><td>Po501(SU)</td><td>Eth</td><td>LACP</td><td>Eth1/1(P)</td></tr></table>	Group	Port-Channel	Type	Protocol	Member Ports	501	Po501(SU)	Eth	LACP	Eth1/1(P)																																				
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The **Topology** view reflects the connectivity of the devices that have been successfully onboarded in the fabric.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>



Configure Host Interface

Now we will configure the server access port that is connected to Site1-Access1.

- Configure **Ethernet1/5** on **Site1-Access1** as an **access port** to provide connectivity for **Server3**

Click the **Connectivity** tab and then select the **Interfaces** sub-tab.

In the filter field, enter **Interface == Ethernet1/5**.

Check the box next to **Access1** and then, click **Actions > Edit Configuration**.

Soweto-LAN-DC Refresh View in topology Actions

Inventory **Connectivity** Segmentation and security Configuration policies Anomalies Advisories Integrations History

Interfaces Interface groups Links Inter-fabric L3 neighbors Endpoints Routes Flows Virtual infrastructure

Interface == Ethernet1/5 Edit Clear All Actions ^

	Interface	Switch	Admin status	Operational status	Reason	Intended configuration mode
☒	Ethernet1/5	Site1-Access1	↑ Up	↑ Up	ok	trunk

Create interface
 Edit configuration
 Configuration >
 Interface group >

Change the interface Mode from Trunk (default) to Access and Enable BPDU guard since this is a Host port where a server will be connected to.

Edit interface(s)

Interface

Site1-Access1 : Ethernet1/5

Mode*

Trunk

Policy*

Host

Policy Options

General Parameters Storm Control

Enable BPDU Guard*

default

Enable spanning-tree bpduguard: true='enable', false='disable', no='return to default settings'

Configure BPDU Filter

default

Configure spanning-tree bpdufilter, no='return to default settings'

Spanning-tree Link-type

auto

Specify a link type for spanning tree protocol use, default is auto

☒ Enable Port Type Fast

Enable spanning-tree edge port behavior

Edit interface(s)

Interface

Site1-Access1 : Ethernet1/5

Mode*

Access

Policy*

Host

Policy Options

General Parameters Storm Control

Enable BPDU Guard*

enable

Enable spanning-tree bpduguard: true='enable', false='disable', no='return to default settings'

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Configure spanning-tree bpdufilter, no='return to default settings'

Spanning-tree Link-type

auto

Specify a link type for spanning tree protocol use, default is auto

☒ Enable Port Type Fast

Enable spanning-tree edge port behavior

Click Deploy to push the configuration to the Access switch.

Deploy interfaces configuration

1

Config preview

2

Deploy progress

Filter by attributes

Fabric name	Device name	Interface	Admin status	Operation status	Pending config
Soweto-LAN-DC	Site1-Access1	Ethernet1/5	↑ Up	↑ Up	12 Lines

Pending config - Soweto-LAN-DC - Ethernet1/5 - Site1-Access1

Pending config Side-by-side comparison

```

1 interface ethernet1/5
2   no spanning-tree port type edge trunk
3   no switchport trunk allowed vlan none
4   no switchport mode trunk
5 interface ethernet1/5
6   switchport
7   switchport mode access
8   mtu 9216
9   spanning-tree bpduguard enable
10  spanning-tree port type edge
11  no shutdown
12 configure terminal

```

The port is now configured as an **access port**.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

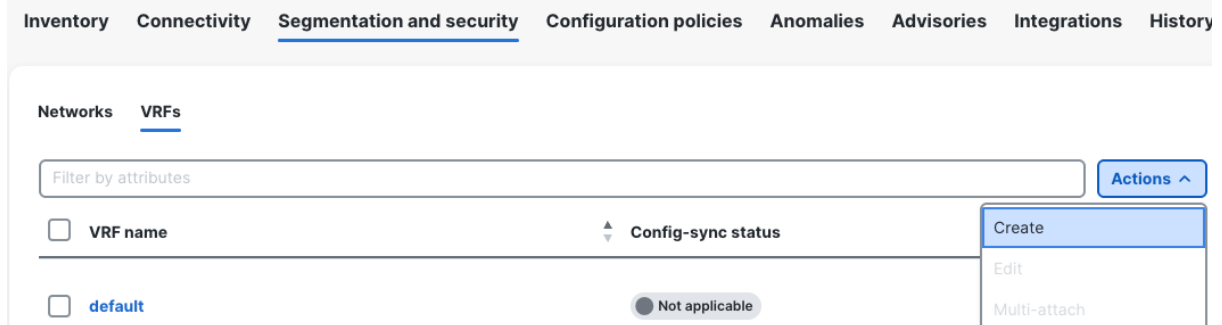
Create a Network and a VRF

We can now begin creating networks and VRFs to enable network connectivity for our endpoints.

- Create **blue_vrf**
- Create the **blue-net3** network within **blue_vrf** for **Server3**

Click the **Segmentation and security** tab.

Next, select the **VRFs** tab, click the white **Actions** button, and then click **Create**.



In the **Create VRF** wizard, enter the VRF name of **blue_vrf**.
Leave all other configuration default, then click **Create**.

Create VRF

VRF name*
blue_vrf

General Parameters Advanced

Routing Protocol
ebgp
VRF Lite Aggregation-Core or Collapsed Core-WAN Peering Protocol (from Fabric Settings)

IP Version*
both
Choice of IPv4, IPv6 or both

OSPF Process Tag
OSPF Routing Process Tag (from Fabric Settings)

OSPF Area Id
OSPF Area Id in IP address format

OSPFv3 Process Tag
OSPFv3 Routing Process Tag (from Fabric Settings)

OSPFv3 Area Id
OSPFv3 Area Id in IP address format

VRF Description

☒ Enable Auto Peering over SVI Between vPC Aggregations
Flag to control per VRF iBGP/OSPF peering between Aggregations, the protocol to use is based on VRF Lite routing protocol in Fabric Settings

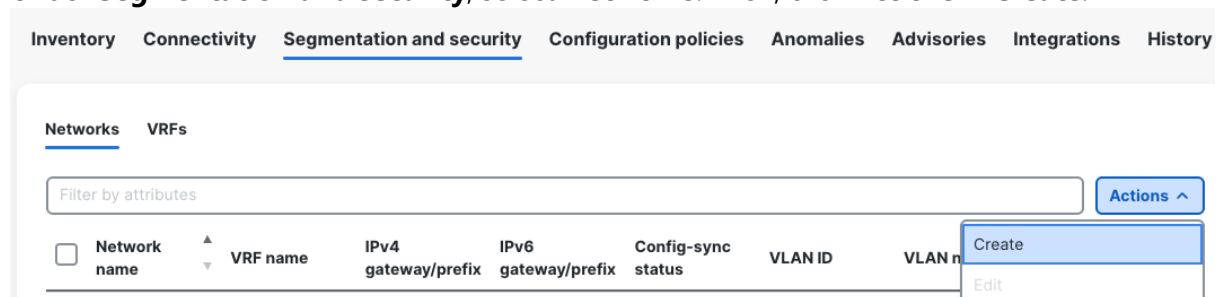
VRF Interface MTU
9216
68-9216

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

The VRF is created and appears in the VRF list.



Under **Segmentation and security**, select **Networks**. Then, click **Actions > Create**.



To create the first network, enter the following information in the **Create Network** menu:

Parameter	Value or Selection
Network Name	blue-net3
Network mode	Layer 3
VRF Name	blue_vrf
Propose VLAN	Click This automatically derives a VLAN from the VLAN pool in the Fabric Resources.
IPv4 Gateway/NetMask	192.168.2.1/24
Interface IPv4 addr on active	192.168.2.2
Interface IPv4 addr on standby	192.168.2.3

Create network

Network name*

blue-net3

Network mode*

Layer 3

VRF name*

blue_vrf

VLAN ID*

2300

[Propose VLAN](#)

General Parameters Advanced

IPv4 Gateway/NetMask

192.168.2.1/24

Example 192.0.2.1/24. Address for FHRP VIP

Interface IPv4 addr on active*

192.168.2.2

Example 192.0.2.2. Interface IP address on the active/master device

Interface IPv4 addr on standby*

192.168.2.3

Example 192.0.2.3. Interface IP address on the standby/backup device

Click **Create**.

Overview **Inventory** **Connectivity** **Segmentation and security** **Configuration policies** **Anomalies**

Networks VRFs

Filter by attributes

☐	Network name	VRF name	IPv4 gateway/prefix	IPv6 gateway/prefix	Config-sync status	VLAN ID
☐	blue-net3	blue_vrf	192.168.2.1/24		Not applicabl	2300

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Attach Network to the Access Port

1. To begin attaching the blue-net3 network to the interface connected to Server3, navigate to the **Segmentation and security > Networks** tab.
2. Click **blue-net3** then click the **Network Attachments** tab. Select the check box next to the **Access1** switch then click **Actions > Edit**.

Network overview - blue-net3 Refresh

Overview Network attachments VRF

Filter by attributes Actions ^

	Network name	VLAN ID	Switch	Interfaces	Configuration status	Attachment
☐	blue-net3		Site1-Agg1	NA	Not applica	Detached
☒	blue-net3		Site1-Access1	NA	Not applica	Detached

History

Edit

Preview

Deploy

In the **Edit Network Attachment** screen, click the slider next to **Attach**. Click **Actions > Add network interface attachment**.

Edit network attachment - blue-net3

Site1-Access1 (905CRZXTNQ4)

☒ Attach

VLAN*
2300 Select VLAN

Attached interfaces for this device

Filter by attributes Actions ^

☐	Interface	Switch	Mode	Customer VLAN
☐	Ethernet1/3	Site1-Access1	trunk	
☐	Ethernet1/4	Site1-Access1	trunk	
☒	Ethernet1/5	Site1-Access1	access	

Add network interface attachment

Select **Ethernet1/5** on **Site1-Access1**. Once selected, click **Add** in the bottom right.

[← Back](#)

Add network interface attachment

Filter by attributes

☒	Interface	Switch	Mode
☐	Ethernet1/3	Site1-Access1	trunk
☐	Ethernet1/4	Site1-Access1	trunk
☒	Ethernet1/5	Site1-Access1	access

Edit network attachment - blue-net3

Site1-Access1 (905CRZXTNQ4)

☒ Attach

VLAN*

2300

Select VLAN

Attached interfaces for this device

Filter by attributes

☐	Interface	Switch	Mode	Customer VLAN	Native VLAN
☐	Ethernet1/5	Site1-Access1	access		No

Click **Save**

Network overview - blue-net3

Overview Network attachments VRF

Filter by attributes

☐	Network name	VLAN ID	Switch	Interfaces	Configuration status	Attachment	Switch role
☐	blue-net3	2300	Site1-Agg1	NA	Pending	Attached	Aggregation
☐	blue-net3	2300	Site1-Access1	Ethernet1/5	Pending	Attached	Access
☐	blue-net3	2300	Site1-Agg2	NA	Pending	Attached	Aggregation

On the **Segmentation and security > Networks** tab. The **Config-sync status** for blue_vrf should be **Pending**.

Overview Inventory Connectivity Segmentation and security Configuration policies Anomalies

Networks VRFs

Filter by attributes

☐	Network name	VRF name	IPv4 gateway/prefix	IPv6 gateway/prefix	Config-sync status	VLAN ID
☐	blue-net3	blue_vrf	192.168.2.1/24		Pending	2300

Review the pending network configuration before deploying.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Deploy configuration - Soweto-LAN-DC

1 Config preview									
2 Deploy progress									
Filter by attributes								Resync all	
Switch name	Role	Configuration sync status	Pending config	Diff	Status description	Progress	Resync switch		
Site1-Access2	Access	In sync	0 Lines	+0 -0	In-Sync		Resync		
Site1-Access3	Access	In sync	0 Lines	+0 -0	In-Sync		Resync		
Site1-Access1	Access	Out of sync	20 Lines	+5 -2	Out-of-Sync		Resync		
Site1-Agg1	Aggregation	Out of sync	55 Lines	+36 -2	Out-of-Sync		Resync		
Site1-Agg2	Aggregation	Out of sync	56 Lines	+37 -2	Out-of-Sync		Resync		

When you review the "Expected config" on the Access switch you will see that the VLAN is created, the access VLAN] added to the access port and the VLAN is allowed on the vPC peer link.

Pending config - Soweto-LAN-DC - Site1-Access1

Pending config Side-by-side comparison

Running config	Expected config
233 hidden lines spanning-tree bpduguard enable spanning-tree port type edge switchport switchport mode access interface ethernet1/50 mtu 9216	233 hidden lines spanning-tree bpduguard enable spanning-tree port type edge switchport switchport access vlan 2300 switchport mode access interface ethernet1/50 mtu 9216
97 hidden lines interface port-channel501 description access-connected-to-vPC-aggregation:Site1-Agg2~Site1-Agg1-vPC1 mtu 9216 spanning-tree bpduguard disable spanning-tree guard loop switchport mode trunk switchport trunk allowed vlan none ip domain-lookup ipv6 switch-packets lla line console	97 hidden lines interface port-channel501 description access-connected-to-vPC-aggregation:Site1-Agg2~Site1-Agg1-vPC1 mtu 9216 no shutdown spanning-tree bpduguard disable spanning-tree guard loop switchport switchport mode trunk switchport trunk allowed vlan 2300 ip domain-lookup ipv6 switch-packets lla line console
7 hidden lines limit-resource port-channel minimum 0 maximum 511 limit-resource vlan minimum 16 maximum 4094 limit-resource vrf minimum 2 maximum 4096 version 10.6(1) Bios:version vlan 1	7 hidden lines limit-resource port-channel minimum 0 maximum 511 limit-resource vlan minimum 16 maximum 4094 limit-resource vrf minimum 2 maximum 4096 vlan 1,2300 version 10.6(1) Bios:version

On the aggregation switches, the SVI is created along with the HSRP based configurations. The aggregation switches also exchange routing information within blue_vrf through an internal BGP session. In this lab, Nexus Dashboard allocates VLAN 2000 for the inter-aggregation peering connection. This serves a different purpose from Server3's VLAN 2300: it supports routing communication between the aggregation peers.

Pending config - Soweto-LAN-DC - Site1-Agg1

Pending config Side-by-side comparison

Running config	Expected config
✱ 15 hidden lines <pre>interface Vlan1 no ip redirects no ipv6 redirects interface ethernet1/1 channel-group 1 mode active mtu 9216</pre>	✱ 15 hidden lines <pre>interface Vlan1 no ip redirects no ipv6 redirects interface Vlan2000 ip address 10.35.0.17/30 ipv6 address 2001::10:35:0:1/126 mtu 9216 no ip redirects no ipv6 redirects no shutdown vrf member blue_vrf interface Vlan2300 hsrp 1 ip 192.168.2.1 preempt hsrp version 2 ip address 192.168.2.3/24 tag 12345 no ip redirects no ipv6 redirects no shutdown vrf member blue_vrf interface ethernet1/1 channel-group 1 mode active mtu 9216</pre>

For every VRF that will be created, an iBGP session will be created. As observed below Vlan2000 is auto-created to enable iBGP sessions between the aggregation switches.

Pending config - Soweto-LAN-DC - Site1-Agg1

Pending config Side-by-side comparison

Running config	Expected config
<pre>interface port-channel1 description vPC-aggregation-connected-to-access:Site1-Access1-port-channel501 mtu 9216 spanning-tree bpduguard disable spanning-tree guard root switchport mode trunk switchport trunk allowed vlan none vpc 1 interface port-channel500 description "vpc-peer-link Site1-Agg1--Site1-Agg2"</pre> ✱ 21 hidden lines	<pre>interface port-channel1 description vPC-aggregation-connected-to-access:Site1-Access1-port-channel501 mtu 9216 no shutdown spanning-tree bpduguard disable spanning-tree guard root switchport switchport mode trunk switchport trunk allowed vlan 2300 vpc 1 interface port-channel500 description "vpc-peer-link Site1-Agg1--Site1-Agg2"</pre> ✱ 21 hidden lines
<pre>match tag 12345 router bgp 65002 router-id 10.2.0.1</pre>	<pre>match tag 12345 router bgp 65002 router-id 10.2.0.1 vrf blue_vrf address-family ipv4 unicast redistribute direct route-map FABRIC-RMAP-REDIST-SUBNET address-family ipv6 unicast redistribute direct route-map FABRIC-RMAP-REDIST-SUBNET neighbor 10.35.0.18 address-family ipv4 unicast send-community both remote-as 65002 neighbor 2001::10:35:0:2 address-family ipv6 unicast send-community both remote-as 65002</pre>

The defined VRF (in this case blue_vrf) will be added to the switches.

<pre> version 10.6(1) Bios:version vlan 1 vpc domain 1 auto-recovery reload-delay 360 delay restore 150 ip arp synchronize ipv6 nd synchronize peer-gateway peer-keepalive destination 198.18.4.130 source 198.18.4.129 peer-switch </pre>	<pre> vlan 1,2000,2300 version 10.6(1) Bios:version vpc domain 1 auto-recovery reload-delay 360 delay restore 150 ip arp synchronize ipv6 nd synchronize peer-gateway peer-keepalive destination 198.18.4.130 source 198.18.4.129 peer-switch vrf context blue_vrf address-family ipv4 unicast address-family ipv6 unicast </pre>
--	---

```

Site1-Agg1# show hsrp brief
*:IPv6 group    #:group belongs to a bundle
                |
                P indicates configured to preempt.
Interface      Grp   Prio P State      Active addr      Standby addr      Group addr
Vlan2300       1    100 P Standby    192.168.2.2      local             192.168.2.1
(conf)

!
Site1-Agg1# show hsrp interface vlan 2300
Vlan2300 - Group 1 (HSRP-V2) (IPv4)
  Local state is Standby, priority 100 (Cfged 100), may preempt
  Forwarding threshold(for vPC), lower: 0 upper: 100
  Hellotime 3 sec, holdtime 10 sec
  Next hello sent in 0.196000 sec(s)
  Virtual IP address is 192.168.2.1 (Cfged)
  Active router is 192.168.2.2, priority 120 expires in 9.185000 sec(s)
  Standby router is local
  Authentication text "cisco"
  Virtual mac address is 0000.0c9f.f001 (Default MAC)
  1 state changes, last state change 00:01:08
  IP redundancy name is hsrp-Vlan2300-1 (default)

```

Verifications

<pre> Site1-Access1# show run int e1/5 !Command: show running-config interface Ethernet1/5 version 10.6(1) Bios:version interface Ethernet1/5 switchport access vlan 2300 spanning-tree port type edge spanning-tree bpduguard enable mtu 9216 </pre>	<pre> Site1-Access1# show run int po501 !Command: show running-config interface port-channel501 version 10.6(1) Bios:version interface port-channel501 description access-connected-to-vPC- aggregation:Site1-Agg2~Site1-Agg1-vPC1 switchport mode trunk switchport trunk allowed vlan 2300 spanning-tree bpduguard disable spanning-tree guard loop mtu 9216 </pre>
--	---

The server is already pre-configured with the following MAC and IP address (outside the scope of this lab)

IP: 192.168.2.30 **MAC:** 5254.0093.e70a

Server should now be able to ping the default gateway

<pre> cisco@Server3:~\$ ping 192.168.2.1 PING 192.168.2.1 (192.168.2.1): 56 data bytes 64 bytes from 192.168.2.1: seq=1 ttl=255 time=7.498 ms 64 bytes from 192.168.2.1: seq=2 ttl=255 time=6.024 ms 64 bytes from 192.168.2.1: seq=3 ttl=255 time=7.019 ms </pre>	<pre> Site1-Access1# show mac address-table vlan 2300 grep 1/5 * 2300 5254.0093.e70a dynamic NA F F Eth1/5 5 </pre>
--	---

64 bytes from 192.168.2.1: seq=4 ttl=255 time=5.565 ms ^C --- 192.168.2.1 ping statistics --- 5 packets transmitted, 4 packets received, 20% packet loss round-trip min/avg/max = 5.565/6.526/7.498 ms	
cisco@Server3:~\$ ifconfig eth1 Link encap:Ethernet HWaddr 52:54:00:93:E7:0A inet addr:192.168.2.30 Bcast:192.168.2.255 Mask:255.255.255.0 inet6 addr: fe80::5054:ff:fe93:e70a/64	Site1-Agg1# show ip arp vrf blue_vrf grep 5254.0093.e70a 192.168.2.30 00:04:03 5254.0093.e70a Vlan2300

Create Core Fabric

This section shows how to create the Core Layer of the Data Center network. The core layer provides external connectivity and can allow multi-site connectivity.

1. In Nexus Dashboard, navigate to Manage > Fabrics > Actions and click Create Fabric.

Select **Create new LAN fabric** and then click **Next**.

Create/Onboard Fabric

What is a fabric?

1 Select a category

Create new LAN fabric

2 Select a type

3 Settings

4

Select a category

Select if you want to create a new fabric or onboard and automate configuration of an existing fabric.

[Learn more](#)

+

Create new LAN fabric

Select this option to provision a new network comprising of Cisco NX-OS, IOS-XE, IOS-XR, and/or 3rd party devices through Nexus Dashboard.

Select the “**External and Inter-Fabric Connectivity**” type.

Select a type

Switches in this fabric will be configured automatically based on the option you choose.

VXLAN

Automate a VXLAN BGP EVPN fabric for Cisco Nexus (NX-OS) and/or Catalyst (IOS-XE) switches.

Classic LAN

Automate the provisioning of a 2 or 3-tier Traditional Classical Ethernet Network.

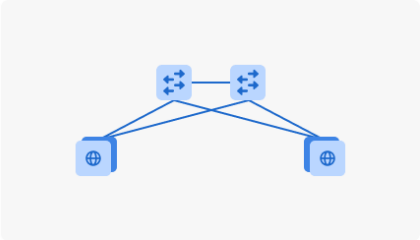
AI

Automate a Nexus (NX-OS) fabric for top performance AI networks using RoCEv2.

External and Inter-Fabric Connectivity

Monitor or manage any architecture that includes Cisco NX-OS, IOS-XE, IOS-XR and/or 3rd party devices. This includes use cases for External connectivity, Inter-fabric Connectivity Networks (such as ISNs for ACI), and Inter-Pod Networks (IPNs).

Routed



Fabric type

External and Inter-Fabric Connectivity

This fabric type supports a variety of use cases. Configurations from the switches will not be auto imported into the controller. This fabric type supports the following use-cases:

- Inter-fabric connectivity networks—including ISNs and IPNs—provide the underlying infrastructure that connects VXLAN EVPN fabrics across multi-site and multi-cloud deployments.
- Flexible deployments to monitor and/or manage networks that include a mix of any Cisco NX-OS, IOS-XE, IOS-XR and/or 3rd party devices.

Leave the remaining fabric settings at their defaults unless the environment requires otherwise.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Create/Onboard Fabric

[What is a fabric?](#)

✓ **Select a category**
Create new LAN fabric

✓ **Select a type**
External and Inter-F

3 **Settings**
Advanced

4 Advanced settin

5 Summary

6 Fabric creation

Settings

These are the recommended settings for configuring the parameters and capabilities of the new fabric.

Configuration mode ⓘ

☐ Default ☒ Advanced

Name *

SOWETO-BACKBONE

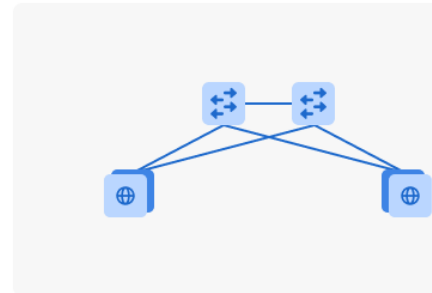
Location *

Johannesburg, ZA

BGP ASN *

65003

1-4294967295 | 1-65535[0-65535]



Fabric type External and Inter-Fabric Connectivity

Uncheck the Fabric Monitor Mode

Create/Onboard Fabric

[What is a fabric?](#)

✓ **Select a category**
Create new LAN fabric

✓ **Select a type**
External and Inter-F

✓ **Settings**
Advanced

4 **Advanced settir**

Advanced settings

The following optional settings will be deployed and/or used when deploying this fabric.

General Parameters Advanced Resources Configuration Backup Bootstrap Flow Monitor Manag

☐ Fabric Monitor Mode

If enabled, fabric is only monitored. No configuration will be deployed

☐ Enable Performance Monitoring (For NX-OS and IOS XE Switches Only)

If enabled, switch metrics are collected through periodic SNMP polling. Alternative to real-time telemetry

Review the **Fabric summary** and then click **Submit**

Create/Onboard Fabric

[What is a fabric?](#)

✓ **Select a category**
Create new LAN fabric

✓ **Select a type**
External and Inter-Fabric Connectivity

✓ **Settings**
Advanced

✓ **Advanced settings**

5 **Summary**

6 Fabric creation

Summary

Review your selections below.

Category

Fabric category New LAN fabric

Type

Fabric type External and Inter-Fabric Connectivity

Fabric sub-type External and Inter-Fabric Connectivity

Settings

Name SOWETO-BACKBONE

Location Johannesburg, ZA

License tier for fabric Premier

Security domain all

BGP ASN 65003

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Add Switches to the Fabric

Create/Onboard Fabric

[What is a fabric?](#)



All done!

The fabric SOWETO-BACKBONE, has successfully been created/onboarded to Nexus Dashboard.

You can modify any of these settings from Manage > Fabrics > Actions by clicking 'Edit fabric settings'

[View fabric details](#) [Add switches to fabric](#) [Create another fabric](#)

Add the Core switch using the same discovery process as before.

Add switches - Fabric: SOWETO-BACKBONE

Switch addition mechanism*
☒ Discover ☐ Move neighbor switches

Seed switch details

Seed IP*

Ex: "2.2.2.20" or "10.10.10.40-60" or "2.2.2.20, 2.2.2.21"

Device type*

Authentication / Privacy*

Username*

Password*
 [Show](#)

☐ Set as individual device write credential

Max hops*

Discover switches

Add switches - Fabric: SOWETO-BACKBONE

Switch addition mechanism*
☒ Discover ☐ Move neighbor switches

Seed switch details

Fabric SOWETO-BACKBONE	Switch 198.18.4.124-125	Authentication protocol md5	Username admin
Password ☐ Set as individual device write credential	Max hops 0	Preserve config ☒ Enabled	

[← Back](#)

Discovery results

Filter by attributes

☐ Switch name	☐ Serial number	☐ IP address	☐ Model	☐ Version	☐ Status	☐
☐ Core1	9RM171U1X07	198.18.4.124	N9K-C9300v	10.6(1)	☒ Manage	
☐ Core2	900JLZX4A1	198.18.4.125	N9K-C9300v	10.6(1)	☒ Manage	

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Add Switches to the fabric and change the device roles to Core Router.

SOWETO-BACKBONE Refresh View in topology Action

Inventory Connectivity Segmentation and security Configuration policies Anomalies Advisories Integrations History

Switches VPC pairs DPUs

Filter by attributes Actions ^

☒	Name	IP address	Model	Config status
☒	Core1	198.18.4.124	N9K-C9300v	Not applicable
☒	Core2	198.18.4.125	N9K-C9300v	Not applicable

- Add switches
- Configuration >
- Discovery >
- Set role

Select **Role** to **Core Router**

Filter by attributes

☒ Name

☒ Core1

☒ Core2

Select Role

Search Role

- Border Super Spine
- Border Gateway Super Spine
- Access
- Aggregation
- Edge Router
- Core Router**

Configuration sync status

Not applicable

Not applicable

Monitor the deployment progress until it completes.

[← Back](#)

Deploy configuration - SOWETO-BACKBONE

1 Config preview 2 Deploy progress

Filter by attributes Resync all

Switch name	IP address	Role	Serial number	Configuration sync status	Pending config	Diff	Status descrip...	Progress
Core1	198.18.4.124	Core Router	9RM171U1X07	Out of sync	5 Lines	+3 -0	Out-of-Sync	
Core2	198.18.4.125	Core Router	900JJLZX4A1	Out of sync	5 Lines	+3 -0	Out-of-Sync	

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Deploy all

Recalculate All Fabrics¹

Nexus Dashboard is capable of comparing the hosts it manages within individual fabrics to the CDP neighbors' output from the neighboring devices

Go to the LAN-DC Site.

1. Navigate to Manage > Fabrics and click Soweto-LAN-DC.
2. In the top-right corner, click the blue Actions button, then click Recalculate and deploy. You should see a pending configuration on the aggregate switches.

This configuration will run 'no shutdown' all interfaces between the fabric and the Core fabric. Notice how the interfaces are automatically changed from trunk port to routed ports.

Click **Deploy all**.

Pending config - Soweto-LAN-DC - Site1-Agg1



Pending config Side-by-side comparison

Running config	Expected config
✱ 169 hidden lines switchport mode trunk switchport trunk allowed vlan none interface ethernet1/4 mtu 9216 switchport mode trunk switchport trunk allowed vlan none interface ethernet1/40 mtu 9216 switchport mode trunk ✱ 35 hidden lines switchport mode trunk switchport trunk allowed vlan none interface ethernet1/5 mtu 9216 switchport mode trunk switchport trunk allowed vlan none interface ethernet1/50 mtu 9216 switchport mode trunk	✱ 169 hidden lines switchport mode trunk switchport trunk allowed vlan none interface ethernet1/4 description connected-to-Core1-Ethernet1/3 mtu 9216 no shutdown no switchport interface ethernet1/40 mtu 9216 switchport mode trunk ✱ 35 hidden lines switchport mode trunk switchport trunk allowed vlan none interface ethernet1/5 description connected-to-Core2-Ethernet1/3 mtu 9216 no shutdown no switchport interface ethernet1/50 mtu 9216 switchport mode trunk

Deploy all

Go to the Backbone fabric (Core).

Recalculate and Deploy.

[← Back](#)

Deploy configuration - SOWETO-BACKBONE

1

2

Config preview

Deploy progress

[Resync all](#)

Switch name	IP address	Role	Serial number	Configuration sync status	Pending config	Diff	Status description	Progress	
Core2	198.18.4.1 25	Core Router	900JLZX4 A1	Out of sync	12 Lines	+8 -0	Out-of-Sync		
Core1	198.18.4.1 24	Core Router	9RM171U1 XO7	Out of sync	12 Lines	+8 -0	Out-of-Sync		

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

Review the pending interface configuration before deploying.

Pending config - SOWETO-BACKBONE - Core1

Pending config Side-by-side comparison

```
interface ethernet1/3
  no switchport
  mtu 9216
  description connected-to-Site1-Agg1-Ethernet1/4
  no shutdown
configure terminal
interface ethernet1/4
  no switchport
  mtu 9216
  description connected-to-Site1-Agg2-Ethernet1/4
  no shutdown
configure terminal
```

Confirm each Core switch has formed a CDP neighborship with both aggregation switches over the links that were just enabled.

Core1# show cdp neighbors	Core2# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge	Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
Repeater, S - Switch, H - Host, I - IGMP, r -	Repeater, S - Switch, H - Host, I - IGMP, r -
Device, V - VoIP-Phone, D - Remotely-Managed-	Device, V - VoIP-Phone, D - Remotely-Managed-
s - Supports-STP-Dispute	s - Supports-STP-Dispute
Device-ID Local Intrfce Hldtme Capability	Device-ID Local Intrfce Hldtme Capability
Platform Port ID	Platform Port ID
Site1-Agg1 (9GTE45OSVS7)	Site1-Agg1 (9GTE45OSVS7)
Eth1/3 128 R S s N9K-	Eth1/3 128 R S s N9K-
C9300v Eth1/4	C9300v Eth1/5
Site1-Agg2 (95NLDHLAI0D)	Site1-Agg2 (95NLDHLAI0D)
Eth1/4 128 R S s N9K-	Eth1/4 128 R S s N9K-
C9300v Eth1/4	C9300v Eth1/5
Total entries displayed: 2	Total entries displayed: 2

Both Core switches see Site1-Agg1 and Site1-Agg2, confirming the physical links to the aggregation layer are up.

VRF Lite Extension

This section shows how to enable VRF-Lite Extension Between the Aggregation and Core Layers. VRF-Lite extends the blue_vrf routing context from the aggregation layer to the core through dedicated Layer 3 connections. In this lab, these connections use sub-interfaces and eBGP to exchange routes

At this stage, VRF Lite extension must be enabled from the LAN fabric.

The VRF Lite interfaces will select IP addresses from the IPv4 VRF Subnet IP Range pool that was specified during creation. This pool is also used for VRFs.

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

In the LAN-DC fabric navigate to **Segmentation and security** > **VRFs** tab then click **blue_vrf**.

Go to **VRF attachments**, select the VRF for both aggregation switches>>**Actions** >>**Edit**.

VRF overview - blue_vrf

Overview VRF attachments Networks

Filter by attributes							History Edit Preview Deploy Import
☒ VRF name	☐ VLAN ID	☐ Switch	☐ Configuration status	☐ Attachment	☐ Switch role		
☒ blue_vrf	2000	Site1-Agg1	Deployed	Attached	Aggregation		
☒ blue_vrf	2000	Site1-Agg2	Deployed	Attached	Aggregation		

Enable the VRF Lite extension for the VRF by changing the Extend option from NONE to VRF-LITE

Edit VRF attachment - blue_vrf

Site1-Agg1(9GTE450SVS7) - Site1-Agg2(95NLDHLA10D)

Detach ☒ Attach

Extend*

NONE

Site1-Agg1(9GTE450SVS7)

CLI Freeform Config

[Edit >](#)

All configs should strictly match the 'show run' output, including cases and new line
Any mismatches will yield unexpected diffs during deploy

VLAN

2000

VLAN Name

if > 32 chars enable:system vlan long-name

SVI IPv4 Address/Netmask

10.35.0.17/30

vPC Peer SVI IPv4 Address

10.35.0.18

SVI IPv6 Address/Netmask

2001::10:35:0:1/126

vPC Peer SVI IPv6 Address

2001::10:35:0:2

Site1-Agg2(95NLDHLA10D)

CLI Freeform Config

[Edit >](#)

All configs should strictly match the 'show run' output, including cases and new line
Any mismatches will yield unexpected diffs during deploy

VLAN

2000

VLAN Name

if > 32 chars enable:system vlan long-name

SVI IPv4 Address/Netmask

10.35.0.18/30

vPC Peer SVI IPv4 Address

10.35.0.17

SVI IPv6 Address/Netmask

2001::10:35:0:2/126

vPC Peer SVI IPv6 Address

2001::10:35:0:1

Select the Core-facing interfaces to attach the VRF Lite extension to.

Site1-Agg1(9GTE45OSVS7) - Site1-Agg2(95NLDHLAI0D)

Detach ☒ Attach

Extend*

VRF_LITE

Extension

Filter by attributes

Action	Attached	Source Switch	Type	Interface Name	Dest. Switch	Dest. Interface	Encapsula... Dot1q VLAN ID	IPv4 Prefix/Mask	IP Tag	Neighbor IPv4 Address	Neighbor ASN
Edit	Detached	Site1-Agg1	VRF_LITE	Ethernet1/4	Core1	Ethernet1/3	2	10.35.0.1/30		10.35.0.2	65003
Edit	Detached	Site1-Agg1	VRF_LITE	Ethernet1/5	Core2	Ethernet1/3	2	10.35.0.6/30		10.35.0.5	65003
Edit	Detached	Site1-Agg2	VRF_LITE	Ethernet1/4	Core1	Ethernet1/4	2	10.35.0.9/30		10.35.0.10	65003
Edit	Detached	Site1-Agg2	VRF_LITE	Ethernet1/5	Core2	Ethernet1/4	2	10.35.0.14/30		10.35.0.13	65003

Click Attach-all and Save

Site1-Agg1(9GTE45OSVS7) - Site1-Agg2(95NLDHLAI0D)

Detach ☒ Attach

Extend*

VRF_LITE

Extension

Filter by attributes

Action	Attached	Source Switch	Type	Interface Name	Dest. Switch	Dest. Interface	Encapsula... Dot1q VLAN ID	IPv4 Prefix/Mask	IP Tag	Neighbor IPv4 Address	Neighbor ASN
Edit	Attached	Site1-Agg1	VRF_LITE	Ethernet1/4	Core1	Ethernet1/3	2	10.35.0.1/30		10.35.0.2	65003
Edit	Attached	Site1-Agg1	VRF_LITE	Ethernet1/5	Core2	Ethernet1/3	2	10.35.0.6/30		10.35.0.5	65003
Edit	Attached	Site1-Agg2	VRF_LITE	Ethernet1/4	Core1	Ethernet1/4	2	10.35.0.9/30		10.35.0.10	65003
Edit	Attached	Site1-Agg2	VRF_LITE	Ethernet1/5	Core2	Ethernet1/4	2	10.35.0.14/30		10.35.0.13	65003

Deploy the configuration

← Back

VRF overview - blue_vrf

[Edit](#)

[Deploy](#)

Overview VRF attachments Networks

Filter by attributes

Actions

☒	VRF name	VLAN ID	Switch	Configuration status	Attachment	Switch role	Fabric name
☒	blue_vrf	2000	Site1-Agg1	Pending	Attached	Aggregation	Soweto-LAN-DC
☒	blue_vrf	2000	Site1-Agg2	Pending	Attached	Aggregation	Soweto-LAN-DC

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

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Deploy configuration - Soweto-LAN-DC

Filter by attributes

VRF name	Fabric name	Serial number	Switch name	IP address	Role	VRF status	Pending config	Progress
blue_vrf	Soweto-LAN-DC	9GTE45OSVS7	Site1-Agg1	198.18.4.129	Aggregation	Out of syn	30 Lines	
blue_vrf	Soweto-LAN-DC	95NLDHLAI0D	Site1-Agg2	198.18.4.130	Aggregation	Out of syn	30 Lines	

Once deployed, the VRF Lite interfaces are configured on the aggregation side.

Navigate to the **Core Fabric** and click the **Recalculate and Deploy** option.

Deploy configuration - SOWETO-BACKBONE

1 Config preview 2 Deploy progress

Filter by attributes

Switch name	IP address	Role	Serial number	Configuration sync status	Pending config	Diff	Status description	Progress	Resync switch
Core2	198.18.4.125	Core Router	900JLZX4A1	Out of sync	31 Lines	+24 -0	Out-of-Sync		Resync
Core1	198.18.4.124	Core Router	9RM171U1X07	Out of sync	31 Lines	+24 -0	Out-of-Sync		Resync

Review the pending sub-interface and BGP configuration before deploying.

Pending config - SOWETO-BACKBONE - Core2

Pending config Side-by-side comparison

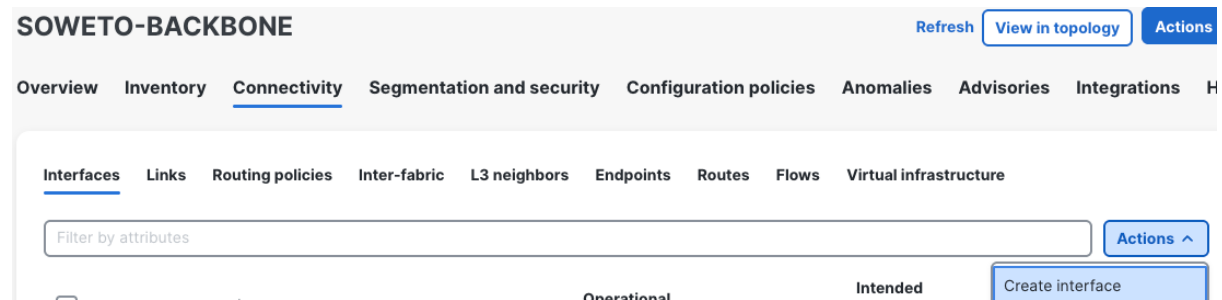
```
vrf context blue_vrf
  address-family ipv4 unicast
exit
router bgp 65003
  vrf blue_vrf
    address-family ipv4 unicast
      neighbor 10.35.0.14
        remote-as 65002
      address-family ipv4 unicast
        send-community both
    exit
  neighbor 10.35.0.6
    remote-as 65002
  address-family ipv4 unicast
    send-community both
configure terminal
interface ethernet1/3.2
  encapsulation dot1q 2
  vrf member blue_vrf
  ip address 10.35.0.5/30
  mtu 9216
  no shutdown
configure terminal
interface ethernet1/4.2
  encapsulation dot1q 2
```

As observed from the Pending configurations, interfaces on the core devices will be configured as sub-interfaces with IP addressing and BGP configurations under the respective VRF will be pushed as well to establish the required BGP neighborship between aggregation and core.

Advertise External Networks from the Core to the LAN

Now the core devices will be configured with loopback IP addresses that will be advertised by BGP to the LAN fabric. The Loopback interfaces on Core1 and Core2 represent external destinations for this lab. Advertising their prefixes through BGP allows the aggregation switches to learn how to reach them.

Navigate to Interfaces under **Connectivity**, Click on the lower Actions and **Create interface**.



The loopbacks are created under interfaces with their respective IP addresses and they are assigned to the relevant VRF.

The image shows two side-by-side 'Create interface' forms. Both forms have the following fields: 'Type' (Loopback), 'Select a device' (Core1 for the left, Core2 for the right), 'Loopback ID' (0), 'Interface VRF' (blue_vrf), and 'Loopback IP' (1.1.1.1 for the left, 2.2.2.2 for the right). The 'Policy Options' section is also visible.

The loopback interfaces are successfully created with a "Base" policy associated to them.

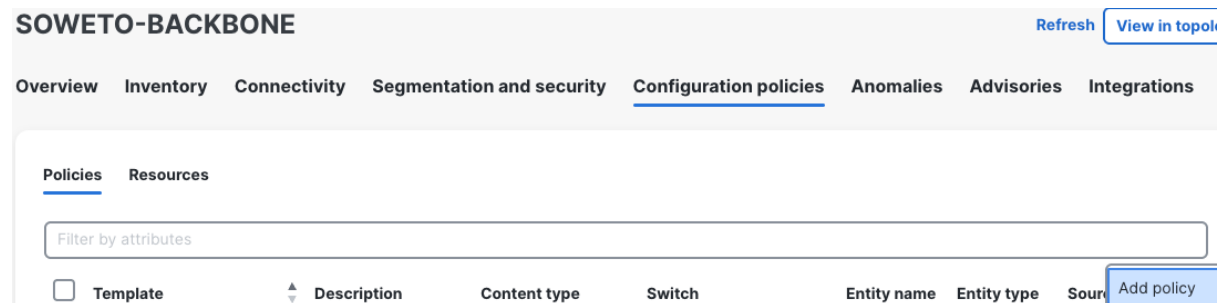
The screenshot shows the SOWETO-BACKBONE network management interface. The top navigation bar includes 'Overview', 'Inventory', 'Connectivity' (selected), 'Segmentation and security', 'Configuration policies', 'Anomalies', and 'Advisories'. Below this, the 'Interfaces' sub-tab is selected. A search bar with 'Interface contains loopback' is visible. The table below shows the status of the created loopback interfaces.

Interface	Switch	Admin status	Operational status	Reason	Intended configuration mode	Policy
Loopback0	Core1	↑ Up	↑ Up	ok	managed	Base
Loopback0	Core2	↑ Up	↑ Up	ok	managed	Base

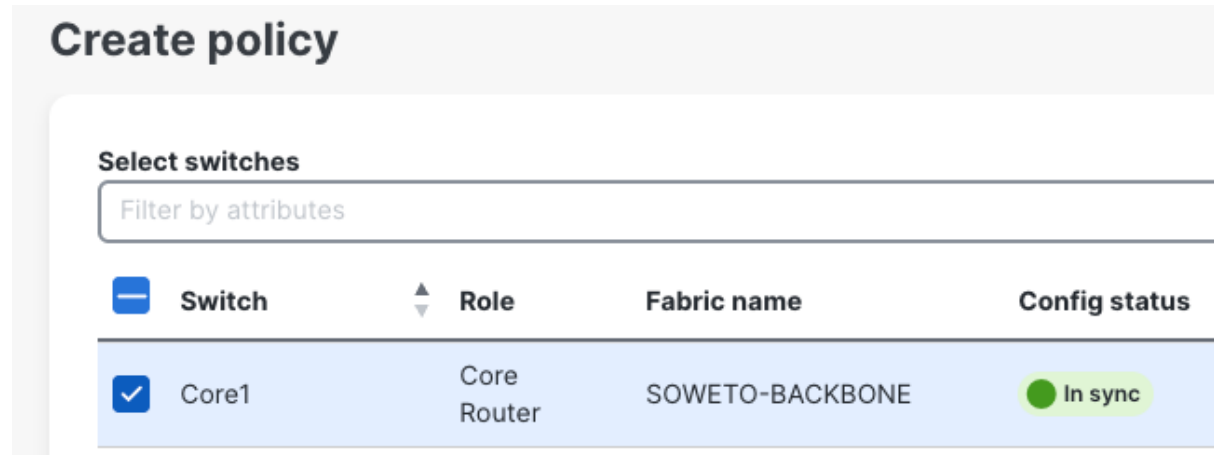
For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

To advertise these networks, the **bgp_vrf_network** policy is required.

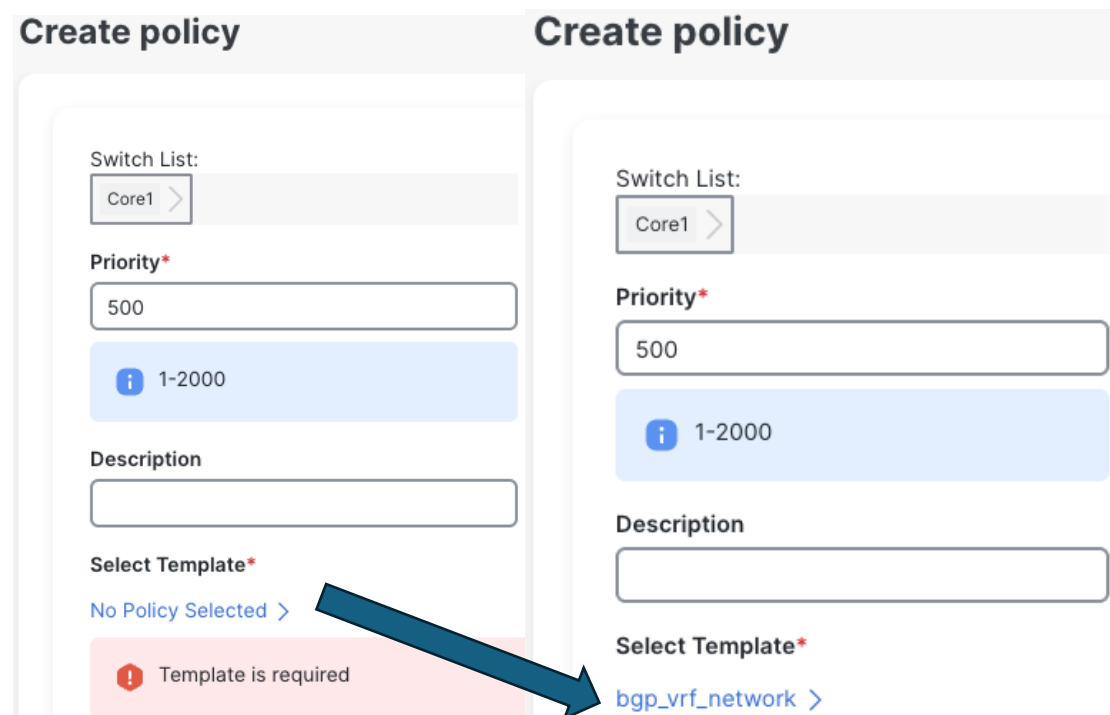
Under the Configuration policies, navigate to Policies, click Actions and Add Policy



Select the Core device and Click Next



Change the Template to “**bgp_vrf_network**”



Put in the BGP AS number (65003), vrf name for the VRF extension along with the IP prefix to advertise, in this case the IP prefix to advertise is the loopback address.

Create policy

Switch List:

Core1 >

Priority*

500

i

1-2000

Description

Select Template*

bgp_vrf_network >

Group

BGP AS #*

65003

BGP Autonomous System number

VRF Name*

blue_vrf

Name of the VRF

IP prefix to advertise*

1.1.1.1/32

IP prefix in CIDR format

The pending configuration shows that a network statement is added to the BGP configuration under the VRF.

Pending config - SOWETO-BACKBONE - Core1

Pending config

Side-by-side comparison

```
router bgp 65003
  vrf blue_vrf
    address-family ipv4 unicast
      network 1.1.1.1/32
configure terminal
```

Repeat the same steps for Core-2 to advertise 2.2.2.2/32.

Verifications

Both Core routers hold their own loopback locally and learn the blue_vrf network (192.168.2.0/24) from the aggregation switches via eBGP.

<pre>Core1# show ip route vrf blue_vrf 1.1.1.1/32, ubest/mbest: 2/0, attached *via 1.1.1.1, Lo0, [0/0], 1w4d, local, tag 12345 *via 1.1.1.1, Lo0, [0/0], 1w4d, direct, tag 12345 192.168.2.0/24, ubest/mbest: 1/0 *via 10.35.0.1, [20/0], 1w4d, bgp-65003, external, tag 65002</pre>	<pre>Core2# show ip route vrf blue_vrf 2.2.2.2/32, ubest/mbest: 2/0, attached *via 2.2.2.2, Lo0, [0/0], 00:49:05, local, tag 12345 *via 2.2.2.2, Lo0, [0/0], 00:49:05, direct, tag 12345 192.168.2.0/24, ubest/mbest: 1/0 *via 10.35.0.14, [20/0], 00:48:53, bgp-65003, external, tag 65002</pre>
--	---

The Core loopbacks (1.1.1.1, 2.2.2.2) are learned via BGP on the aggregation switches.

<pre>Site1-Agg1# show ip route vrf blue_vrf 1.1.1.1/32, ubest/mbest: 1/0 *via 10.35.0.2, [20/0], 00:20:40, bgp-65002, external, tag 65003 2.2.2.2/32, ubest/mbest: 1/0 *via 10.35.0.5, [20/0], 00:01:13, bgp-65002, external, tag 65003 192.168.2.0/24, ubest/mbest: 1/0, attached *via 192.168.2.3, Vlan2300, [0/0], 1w4d, direct, tag 12345</pre>	<pre>Site1-Agg2# show ip route vrf blue_vrf 1.1.1.1/32, ubest/mbest: 1/0 *via 10.35.0.10, [20/0], 1w4d, bgp-65002, external, tag 65003 2.2.2.2/32, ubest/mbest: 1/0 *via 10.35.0.13, [20/0], 00:47:52, bgp-65002, external, tag 65003 192.168.2.0/24, ubest/mbest: 1/0, attached *via 192.168.2.2, Vlan2300, [0/0], 1w4d, direct, tag 12345</pre>
---	---

The server connected to the Access switch is able to ping the external IP addresses.

<pre>cisco@Server3:~\$ ping 1.1.1.1 -c 4 PING 1.1.1.1 (1.1.1.1): 56 data bytes 64 bytes from 1.1.1.1: seq=0 ttl=254 time=8.843 ms 64 bytes from 1.1.1.1: seq=1 ttl=254 time=8.513 ms 64 bytes from 1.1.1.1: seq=2 ttl=254 time=9.732 ms 64 bytes from 1.1.1.1: seq=3 ttl=254 time=7.508 ms --- 1.1.1.1 ping statistics --- 4 packets transmitted, 4 packets received, 0% packet loss</pre>	<pre>cisco@Server3:~\$ ping 2.2.2.2 -c 4 PING 2.2.2.2 (2.2.2.2): 56 data bytes 64 bytes from 2.2.2.2: seq=0 ttl=254 time=8.170 ms 64 bytes from 2.2.2.2: seq=1 ttl=254 time=7.035 ms 64 bytes from 2.2.2.2: seq=2 ttl=254 time=7.000 ms 64 bytes from 2.2.2.2: seq=3 ttl=254 time=8.494 ms --- 2.2.2.2 ping statistics --- 4 packets transmitted, 4 packets received, 0% packet loss</pre>
---	---

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>

References

https://www.cisco.com/c/en/us/td/docs/dcn/ndfc/1222/articles/ndfc-enhanced-classic-lan/enhanced-classic-lan.html#_creating_an_enhanced_classic_lan_fabric

Cisco Live:

Automate Traditional Three-Tier vPC-Based Deployment Using NDFC - BRKDCN-1659

Cisco dcloud:

<https://www.ciscodcloud.com/demo-guide/dcloud/datacenter/cisco-nexus-dashboard-4-2-ec/>

For more labs visit my GitHub repo: <https://github.com/TitusM/Cisco-Data-Center>